JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 23rd January Morning Shift

Question Paper with Answer Key

75	ENGLISH	LIGHT
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Given below are two statements:

Statement I: In Lassaigne's test, the covalent organic molecules are transformed into ionic compounds.

Statement II: The sodium fusion extract of an organic compound having N and S gives prussian blue colour with $FeSO_4$ and $Na_4[Fe(CN)_6]$

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q2. MCQ (Single Correct Answer)

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Match the List - I with List - II

	List - I (Classification of molecules based on octet rule)		**List - II (Example)**
(A)	Molecules obeying octet rule	(I)	NO, NO_2
(B)	Molecules with incomplete octet	(II)	$BCl_3, AlCl_3$
(C)	Molecules with incomplete octet with odd electron	(III)	H_2SO_4, PCl_5
(D)	Molecules with expanded octet	(IV)	CCl_4, CO_2

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-II, B-IV, C-III, D-I

(C) A-IV, B-II, C-I, D-III

(D) A-III, B-II, C-I, D-IV

Q3. MCQ (Single Correct Answer)

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What amount of bromine will be required to convert 2 g of phenol into 2,4,6-tribromophenol?

(Given molar mass in $gmol^{-1}$ of C, H, O, Br are $12, 1, 16, 80$ respectively)

(A) 6.0 g **(B)** 10.22 g **(C)** 20.44 g **(D)** 4.0 g

Q4. MCQ (Single Correct Answer)

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The correct set of ions (aqueous solution) with same colour from the following is:

(A) V^{2+}, Cr^{3+}, Mn^{3+}

(B) Ti^{4+}, V^{4+}, Mn^{2+}

(C) $Sc^{3+}, Ti^{3+}, Cr^{2+}$

(D) Zn^{2+}, V^{3+}, Fe^{3+}

Q5. MCQ (Single Correct Answer)

$$\mathrm{FeO_4^{2-}} \xrightarrow{+2.0\ \mathrm{V}} \mathrm{Fe^{3+}} \xrightarrow{0.8\ \mathrm{V}} \mathrm{Fe^{2+}} \xrightarrow{-0.5\ \mathrm{V}} \mathrm{Fe^{0}}$$

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In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $\mathrm{E}_{\mathrm{FeO_4^{2-}/Fe^{2+}}}$ is :

(A) 1.2 V **(B)** 2.1 V **(C)** 1.4 V **(D)** 1.7 V

Q6. MCQ (Single Correct Answer)

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$\mathrm{CrCl_3 \cdot xNH_3}$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of $0.558^{\circ}\mathrm{C}$. Assuming 100% ionisation of this complex and coordination number of Cr is 6 , the complex will be (Given $\mathrm{K_f = 1.86\ K\ kg\ mol^{-1}}$)

(A) $\mathrm{[Cr(NH_3)_5Cl]Cl_2}$ **(B)** $\mathrm{[Cr(NH_3)_4Cl_2]Cl}$

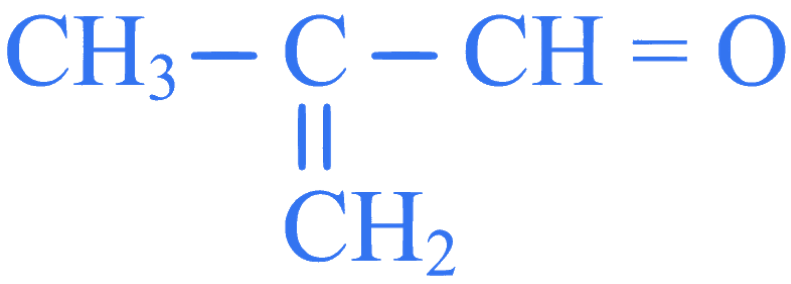
(C) $\mathrm{[Cr(NH_3)_6]Cl_3}$ **(D)** $\mathrm{[Cr(NH_3)_3Cl_3]}$

Q7. MCQ (Single Correct Answer)

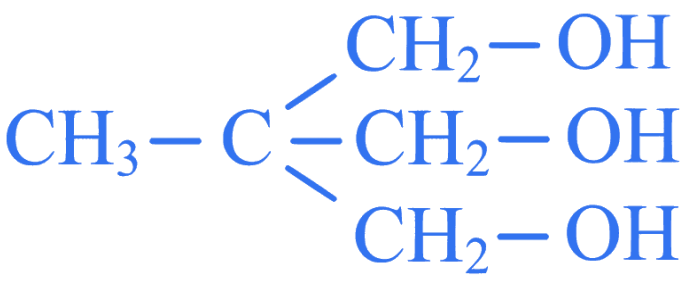
The major product of the following reaction is :

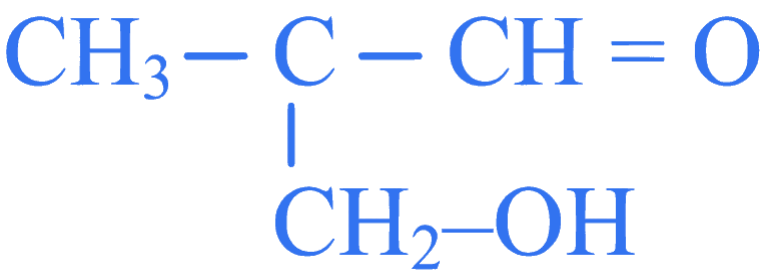
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$$CH_3CH_2CH=O \xrightarrow[\text{reflux}]{\text{excess HCHO, alkali}} ?$$

(A) 

(B) $CH_3-CH_2-CH_2-OH$

(C) 

(D) 

Q8. MCQ (Single Correct Answer)

Propane molecule on chlorination under photochemical condition gives two di-chloro products, " x " and " y ". Amongst " x " and " y ", " x " is an optically active molecule. How many tri-chloro products (consider only structural isomers) will be obtained from " x " when it is further treated with chlorine under the photochemical condition?

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(A) 4 (B) 2 (C) 3 (D) 5

Q9. MCQ (Single Correct Answer)

The incorrect statement among the following is

(A) PH_3 shows lower proton affinity than NH_3.

(B) SO_2 can act as an oxidizing agent, but not as a reducing agent.

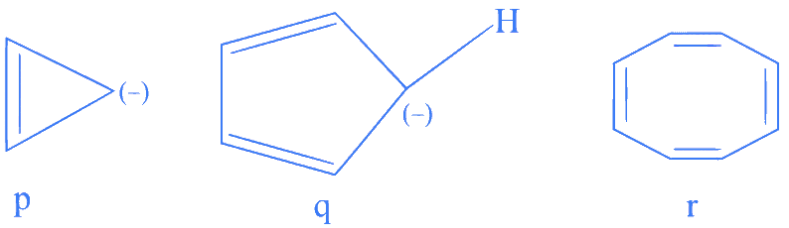
(C) NO_2 can dimerise easily.

(D) PF_3 exists but NF_5 does not.

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Q10. MCQ (Single Correct Answer)

The correct stability order of the following species/molecules is :



(A) $q > r > p$ (B) $q > p > r$ (C) $p > q > r$ (D) $r > q > p$

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Q11. MCQ (Single Correct Answer)

2.8×10^{-3} mol of CO_2 is left after removing 10^{21} molecules from its ' x ' mg sample. The mass of CO_2 taken initially is Given: $N_A = 6.02 \times 10^{23}\ mol^{-1}$

(A) 98.3 mg (B) 196.2 mg (C) 150.4 mg (D) 48.2 mg

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Q12. MCQ (Single Correct Answer)

Given below are two statements:

Statement I: Fructose does not contain an aldehydic group but still reduces Tollen's reagent

Statement II: In the presence of base, fructose undergoes rearrangement to give glucose.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

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Q13. MCQ (Single Correct Answer)

Which of the following happens when NH_4OH is added gradually to the solution containing 1 M A^{2+} and $1MB^{3+}$ ions?

Given : $K_{sp}\ [A(OH)_2] = 9 \times 10^{-10}$ and $K_{sp}\ [B(OH)_3] = 27 \times 10^{-18}$ at 298 K.

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(A) $A(OH)_2$ will precipitate before $B(OH)_3$

(B) $A(OH)_2$ and $B(OH)_3$ will precipitate together

(C) Both $A(OH)_2$ and $B(OH)_3$ do not show precipitation with NH_4OH

(D) $B(OH)_3$ will precipitate before $A(OH)_2$

Q14. MCQ (Single Correct Answer)

Ice at $-5^\circ C$ is heated to become vapor with temperature of $110^\circ C$ at atmospheric pressure. The entropy change associated with this process can be obtained from

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(A) $\int_{268\,K}^{273\,K} C_{p,m}dT + \frac{\Delta H_{m,\text{ fusion}}}{T_f} + \frac{\Delta H_{m,\text{ vaporisation}}}{T_b} + \int_{273\,K}^{373\,K} C_{p,m}dT + \int_{373\,K}^{383\,K} C_{p,m}dT$

(B) $\int_{268\,K}^{383\,K} C_p dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

(C) $\int_{268\,K}^{383\,K} C_p dT + \frac{q_{rev}}{T}$

(D) $\int_{268\,K}^{273\,K} \frac{C_{p,m}}{T}dT + \frac{\Delta H_{m,\text{ fusion}}}{T_f} + \frac{\Delta H_{m,\text{ vaporisation}}}{T_b} + \int_{273\,K}^{373\,K} \frac{C_{p,m}dT}{T} + \int_{373\,K}^{383\,K} \frac{C_{p,m}dT}{T}$

Q15. MCQ (Single Correct Answer)

Match the List - I with List - II

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	List - I Name reaction		**List - II Product obtainable**
(A)	Swarts reaction	(I)	Ethyl benzene
(B)	Sandmeyer's reaction	(II)	Ethyl iodide
(C)	Wurtz Fittig reaction	(III)	Cyanobenzene
(D)	Finkelstein reaction	(IV)	Ethyl fluoride

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-I, C-III, D-IV

(C) A-IV, B-I, C-III, D-II

(D) A-IV, B-III, C-I, D-II

Q16. MCQ (Single Correct Answer)

The element that does not belong to the same period of the remaining elements (modern periodic table) is:

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(A) Platinum (B) Osmium (C) Iridium (D) Palladium

Q17. MCQ (Single Correct Answer)

The d-electronic configuration of an octahedral Co (II) complex having magnetic moment of 3.95 BM is:

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(A) $e^4t_2^3$

(B) $\mathrm{t_{2g}^6e_g^1}$

(C) $\mathrm{t_{2g}^5e_g^2}$

(D) $t_{2g}^3e_g^0$

Q18. MCQ (Single Correct Answer)

The complex that shows Facial - Meridional isomerism is :

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(A) $[\mathrm{Co(en)_2Cl_2}]^+$

(B) $[\mathrm{Co(NH_3)_3Cl_3}]$

(C) $[\mathrm{Co(NH_3)_4Cl_2}]^+$

(D) $[\mathrm{Co(en)_3}]^{3+}$

Q19. MCQ (Single Correct Answer)

Heat treatment of muscular pain involves radiation of wavelength of about 900 nm . Which spectral line of H atom is suitable for this?

Given : Rydberg constant $\mathrm{R_H} = 10^5\ \mathrm{cm^{-1}}$, $\mathrm{h} = 6.6 \times 10^{-34}\ \mathrm{J\,s}, \mathrm{c} = 3 \times 10^8\ \mathrm{m/s}\Big)$

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(A) Lyman series, $\infty \rightarrow 1$

(B) Balmer series, $\infty \rightarrow 2$

(C) Paschen series, $5 \rightarrow 3$

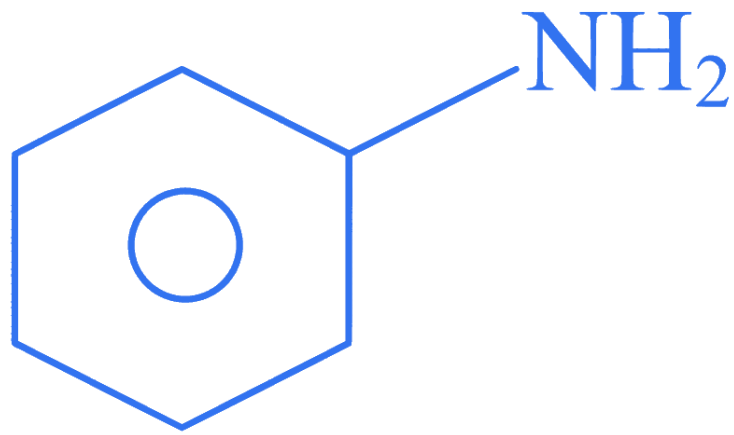
(D) Paschen series, $\infty \rightarrow 3$

Q20. MCQ (Single Correct Answer)

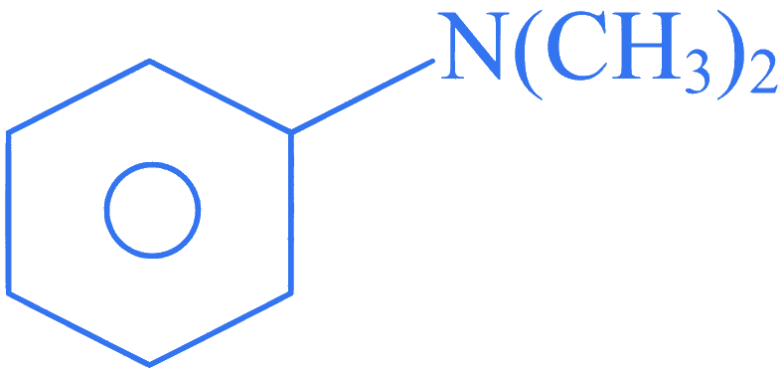
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Which among the following react with Hinsberg's reagent?

A.



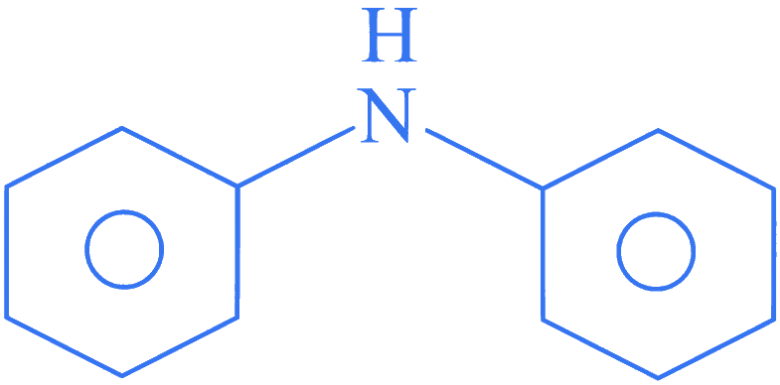
B.



C. $CH_3 - NH_2$

D. $N(CH_3)_3$

E.



Choose the correct answer from the options given below :

(A) A, C and E only **(B)** A, B and E only **(C)** C and D only **(D)** B and D only

Q21. Numerical

For the thermal decomposition of $N_2O_5(g)$ at constant volume, the following table can be formed, for the reaction mentioned below.

$2\,N_2O_5(g) \rightarrow 2\,N_2O_4(g) + O_2(g)$

Sr. No.	Time/s	Total pressure/(atm)
1	0	0.6
2	100	'x'

$x =$ _____ $\times 10^{-3}$ atm [nearest integer]

Given : Rate constant for the reaction is $4.606 \times 10^{-2}\,s^{-1}$.

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Q22. Numerical

The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are $55.0\,kJ\,mol^{-1}$ and $175.0\,J/K/mol$ respectively. The standard free energy change for this reaction at $25^\circ C$ in J mol^{-1} is _____ (Nearest integer)

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Q23. Numerical

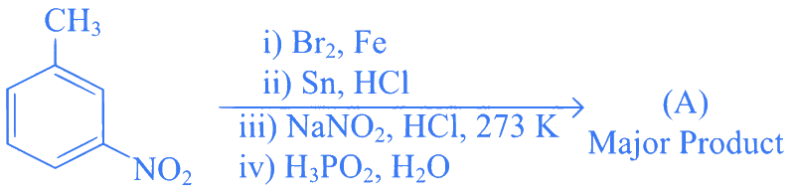
If 1 mM solution of ethylamine produces $pH = 9$, then the ionization constant (K_b) of ethylamine is 10^{-x}. The value of x is _____ (nearest integer).

[The degree of ionization of ethylamine can be neglected with respect to unity.]

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Q24. Numerical

Consider the following sequence of reactions to produce major product (A)



Molar mass of product (A) is _____ $g\,mol^{-1}$.

(Given molar mass in $g\,mol^{-1}$ of C : 12, H : 1, O : 16, Br : 80, N : 14, P : 31)

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Q25. Numerical

During " S " estimation, 160 mg of an organic compound gives 466 mg of barium sulphate. The percentage of Sulphur in the given compound is _____ %.

(Given molar mass in $g\,mol^{-1}$ of $Ba : 137,\ S : 32, O : 16$)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let a curve $y = f(x)$ pass through the points $(0, 5)$ and $(\log_e 2, k)$. If the curve satisfies the differential equation $2(3 + y)e^{2x}dx - \left(7 + e^{2x}\right)dy = 0$, then k is equal to

(A) 32 (B) 8 (C) 4 (D) 16

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Q27. MCQ (Single Correct Answer)

Let P be the foot of the perpendicular from the point $\mathrm{Q}(10, -3, -1)$ on the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z+1}{-2}$. Then the area of the right angled triangle PQR, where R is the point $(3, -2, 1)$, is

(A) $\sqrt{30}$ (B) $9\sqrt{15}$ (C) $3\sqrt{30}$ (D) $8\sqrt{15}$

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Q28. MCQ (Single Correct Answer)

The number of words, which can be formed using all the letters of the word "DAUGHTER", so that all the vowels never come together, is :

(A) 34000 (B) 37000 (C) 35000 (D) 36000

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Q29. MCQ (Single Correct Answer)

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then the domain of $f \circ g$ is

(A) $(0, \infty)$ (B) $[1, \infty)$ (C) $\mathbb{R}$ (D) $[0, \infty)$

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Q30. MCQ (Single Correct Answer)

Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}, \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 2\hat{k}$ and $2\hat{i} + \hat{j} - \hat{k}$ respectively. The altitude from the vertex D to the opposite face ABC meets the median line segment through A of the triangle ABC at the point E. If the length of AD is $\frac{\sqrt{110}}{3}$ and the volume of the tetrahedron is $\frac{\sqrt{805}}{6\sqrt{2}}$, then the position vector of E is

(A) $\frac{1}{6}(7\hat{\mathrm{i}} + 12\hat{\mathrm{j}} + \hat{\mathrm{k}})$ (B) $\frac{1}{12}(7\hat{\mathrm{i}} + 4\hat{\mathrm{j}} + 3\hat{\mathrm{k}})$

(C) $\frac{1}{6}(12\hat{i} + 12\hat{j} + \hat{k})$ (D) $\frac{1}{2}(\hat{i} + 4\hat{j} + 7\hat{k})$

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Q31. MCQ (Single Correct Answer)

If $\frac{\pi}{2} \leq x \leq \frac{3\pi}{4}$, then $\cos^{-1}\left(\frac{12}{13}\cos x + \frac{5}{13}\sin x\right)$ is equal to

(A) $x + \tan^{-1}\frac{5}{12}$ (B) $x - \tan^{-1}\frac{4}{3}$ (C) $x + \tan^{-1}\frac{4}{5}$ (D) $x - \tan^{-1}\frac{5}{12}$

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Q32. MCQ (Single Correct Answer)

One die has two faces marked 1 , two faces marked 2 , one face marked 3 and one face marked 4 . Another die has one face marked 1 , two faces marked 2 , two faces marked 3 and one face marked 4. The probability of getting the sum of numbers to be 4 or 5 , when both the dice are thrown together, is

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(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{4}{9}$ (D) $\frac{1}{2}$

Q33. MCQ (Single Correct Answer)

The value of $(\sin 70^\circ)\left(\cot 10^\circ \cot 70^\circ - 1\right)$ is

(A) 0 (B) 2/3 (C) 1 (D) 3/2

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Q34. MCQ (Single Correct Answer)

If the function

$$f(x) = \begin{cases} \frac{2}{x}\{\sin(k_1+1)x + \sin(k_2-1)x\}, & x < 0 \\ 4, & x = 0 \\ \frac{2}{x}\log_e\left(\frac{2+k_1x}{2+k_2x}\right), & x > 0 \end{cases}$$

is continuous at $x = 0$, then $k_1^2 + k_2^2$ is equal to :

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(A) 5 (B) 10 (C) 20 (D) 8

Q35. MCQ (Single Correct Answer)

If the system of equations

$$(\lambda-1)x + (\lambda-4)y + \lambda z = 5$$
$$\lambda x + (\lambda-1)y + (\lambda-4)z = 7$$
$$(\lambda+1)x + (\lambda+2)y - (\lambda+2)z = 9$$

has infinitely many solutions, then $\lambda^2 + \lambda$ is equal to

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(A) 20 (B) 10 (C) 6 (D) 12

Q36. MCQ (Single Correct Answer)

The value of $\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}} + e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right)dx$ is

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(A) 1 (B) $\log_e 2$ (C) e^2 (D) 2

Q37. MCQ (Single Correct Answer)

Marks obtains by all the students of class 12 are presented in a frequency distribution with classes of equal width. Let the median of this grouped data be 14 with median class interval 12-18 and median class frequency 12. If the number of students whose marks are less than 12 is 18 , then the total number of students is :

(A) 52 (B) 44 (C) 40 (D) 48

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Q38. MCQ (Single Correct Answer)

Let $\left|\frac{\bar{z}-i}{2\bar{z}+i}\right| = \frac{1}{3}, z \in C$, be the equation of a circle with center at C. If the area of the triangle, whose vertices are at the points $(0,0), C$ and $(\alpha, 0)$ is 11 square units, then α^2 equals:

(A) $\frac{121}{25}$ (B) 100 (C) $\frac{81}{25}$ (D) 50

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Q39. MCQ (Single Correct Answer)

Let $\mathrm{I}(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$. If $\mathrm{I}(37) - \mathrm{I}(24) = \frac{1}{4}\left(\frac{1}{\mathrm{b}^{\frac{1}{13}}} - \frac{1}{\mathrm{c}^{\frac{1}{13}}}\right), \mathrm{b}, \mathrm{c} \in \mathcal{N}$, then $3(\mathrm{b}+\mathrm{c})$ is equal to

(A) 39 (B) 22 (C) 40 (D) 26

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Q40. MCQ (Single Correct Answer)

If the line $3x - 2y + 12 = 0$ intersects the parabola $4y = 3x^2$ at the points A and B, then at the vertex of the parabola, the line segment AB subtends an angle equal to

(A) $\frac{\pi}{2} - \tan^{-1}\left(\frac{3}{2}\right)$ (B) $\tan^{-1}\left(\frac{9}{7}\right)$
(C) $\tan^{-1}\left(\frac{11}{9}\right)$ (D) $\tan^{-1}\left(\frac{4}{5}\right)$

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Q41. MCQ (Single Correct Answer)

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to

(A) -120 (B) -1200 (C) -1080 (D) -1020

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Q42. MCQ (Single Correct Answer)

Let the arc AC of a circle subtend a right angle at the centre O. If the point B on the arc AC, divides the arc AC such that $\frac{\text{length of arc } AB}{\text{length of arc } BC} = \frac{1}{5}$, and $\overrightarrow{OC} = \alpha\overrightarrow{OA} + \beta\overrightarrow{OB}$, then $\alpha + \sqrt{2}(\sqrt{3}-1)\beta$ is equal to

(A) $2\sqrt{3}$ (B) $5\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $2-\sqrt{3}$

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Q43. MCQ (Single Correct Answer)

Let the area of a $\triangle PQR$ with vertices $P(5,4), Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O\left(2, \frac{14}{5}\right)$ and $C(c,d)$ respectively, then $c+2d$ is equal to

(A) 3 (B) $\frac{7}{3}$ (C) 2 (D) $\frac{8}{3}$

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Q44. MCQ (Single Correct Answer)

If $\mathrm{A}, \mathrm{B}, \text{and } \left(\text{adj } \left(\mathrm{A}^{-1}\right)+\text{adj } \left(\mathrm{B}^{-1}\right)\right)$ are non-singular matrices of same order, then the inverse of $A\left(\text{adj } \left(A^{-1}\right)+\text{adj } \left(B^{-1}\right)\right)^{-1} B$, is equal to

(A) $\frac{AB^{-1}}{|A|}+\frac{BA^{-1}}{|B|}$

(B) $\text{adj } \left(\mathrm{B}^{-1}\right)+\text{adj } \left(\mathrm{A}^{-1}\right)$

(C) $\mathrm{AB}^{-1}+\mathrm{A}^{-1} \mathrm{B}$

(D) $\frac{1}{|AB|}(\text{adj}(B)+\text{adj}(A))$

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Q45. MCQ (Single Correct Answer)

Let $\mathrm{R}=\{(1,2),(2,3),(3,3)\}$ be a relation defined on the set $\{1,2,3,4\}$. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:

(A) 9 (B) 8 (C) 7 (D) 10

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Q46. Numerical

The sum of all rational terms in the expansion of $\left(1+2^{1/3}+3^{1/2}\right)^6$ is equal to _____.

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Q47. Numerical

If the set of all values of a, for which the equation $5x^3-15x-a=0$ has three distinct real roots, is the interval (α, β), then $\beta-2\alpha$ is equal to _____.

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Q48. Numerical

If the equation $\mathrm{a}(\mathrm{b}-\mathrm{c})\mathrm{x}^2+\mathrm{b}(\mathrm{c}-\mathrm{a})\mathrm{x}+\mathrm{c}(\mathrm{a}-\mathrm{b})=0$ has equal roots, where $\mathrm{a}+\mathrm{c}=15$ and $\mathrm{b}=\frac{36}{5}$, then a^2+c^2 is equal to _____

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Q49. Numerical

Let the circle C touch the line $x-y+1=0$, have the centre on the positive x-axis, and cut off a chord of length $\frac{4}{\sqrt{13}}$ along the line $-3x+2y=1$. Let H be the hyperbola $\frac{x^2}{\alpha^2}-\frac{y^2}{\beta^2}=1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C. Then $2\alpha^2+3\beta^2$ is equal to _____.

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Q50. Numerical

If the area of the larger portion bounded between the curves $x^2+y^2=25$ and $\mathrm{y}=|\mathrm{x}-1|$ is $\frac{1}{4}(\mathrm{b}\pi+\mathrm{c}), \mathrm{b}, \mathrm{c} \in N$, then $\mathrm{b}+\mathrm{c}$ is equal to _____

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

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Given below are two statements:

Statement I: The hot water flows faster than cold water

Statement II: Soap water has higher surface tension as compared to fresh water. In the light above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

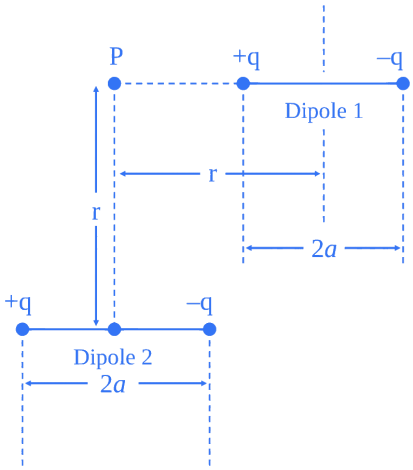
(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q52. MCQ (Single Correct Answer)

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A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r. The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?



(A) $\frac{a}{r} \sim 0.5$ (B) $\frac{a}{r} \sim 3$ (C) $\frac{a}{r} \sim 10$ (D) $\frac{a}{r} \sim 20$

Q53. MCQ (Single Correct Answer)

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What is the lateral shift of a ray refracted through a parallel-sided glass slab of thickness ' h ' in terms of the angle of incidence ' i ' and angle of refraction ' r ', if the glass slab is placed in air medium?

(A) h

(B) $\frac{h\cos(i-r)}{\sin r}$

(C) $\frac{\mathrm{h}\tan(\mathrm{i}-\mathrm{r})}{\tan \mathrm{r}}$

(D) $\frac{h\sin(i-r)}{\cos r}$

Q54. MCQ (Single Correct Answer)

A sub-atomic particle of mass $10^{-30}\ \mathrm{kg}$ is moving with a velocity $2.21 \times 10^{6}\ \mathrm{m/s}$. Under the matter wave consideration, the particle will behave closely like $\left(\mathrm{h} = 6.63 \times 10^{-34}\ \mathrm{J.s}\right)$

(A) X-rays
(B) Infra-red radiation
(C) Gamma rays
(D) Visible radiation

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Q55. MCQ (Single Correct Answer)

The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is

(A) L
(B) $\mathrm{L}^3\,\mathrm{T}^{-2}$
(C) $\mathrm{L}^2\,\mathrm{T}^{-2}$
(D) L^2

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Q56. MCQ (Single Correct Answer)

Match the List - I with List - II

	List - I		List - II
(A)	Pressure varies inversely with volume of an ideal gas.	(I)	Adiabatic process
(B)	Heat absorbed goes partly to increase internal energy and partly to do work.	(II)	Isochoric process
(C)	Heat is neither absorbed nor released by a system.	(III)	Isothermal process
(D)	No work is done on or by a gas.	(IV)	Isobaric process

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV
(B) A-I, B-IV, C-II, D-III
(C) A-III, B-I, C-IV, D-II
(D) A-III, B-IV, C-I, D-II

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Q57. MCQ (Single Correct Answer)

A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J , then the mass of the bullet is _____ grams. (Latent heat of fusion of lead $= 2.5 \times 10^{4}\mathrm{JKg}^{-1}$ and specific heat capacity of lead $= 125\mathrm{JKg}^{-1}\,\mathrm{K}^{-1}$)

(A) 20
(B) 15
(C) 5
(D) 10

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Q58. MCQ (Single Correct Answer)

A solid sphere of mass ' m ' and radius ' r ' is allowed to roll without slipping from the highest point of an inclined plane of length ' L ' and makes an angle 30° with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45° while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio $v_1^2 : v_2^2$ is

(A) $1:3$
(B) $1:2$
(C) $1:\sqrt{2}$
(D) $1:\sqrt{3}$

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Q59. MCQ (Single Correct Answer)

A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1. If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1, after one half-life of n_1 ?

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(A) 1/4 (B) 1/8 (C) 4 (D) 8

Q60. MCQ (Single Correct Answer)

A light hollow cube of side length 10 cm and mass 10 g , is floating in water. It is pushed down and released to execute simple harmonic oscillations. The time period of oscillations is $y\pi \times 10^{-2}\ \mathrm{s}$, where the value of y is (Acceleration due to gravity, $g = 10\ \mathrm{m/s^2}$, density of water $= 10^3\ \mathrm{kg/m^3}$)

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(A) 2 (B) 4 (C) 1 (D) 6

Q61. MCQ (Single Correct Answer)

A spherical surface of radius of curvature R, separates air from glass (refractive index $= 1.5$). The centre of curvature is in the glass medium. A point object ' O ' placed in air on the optic axis of the surface, so that its real image is formed at 'I' inside glass. The line OI intersects the spherical surface at P and $PO = PI$. The distance PO equals to

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(A) 5R (B) 2R (C) 1.5R (D) 3R

Q62. MCQ (Single Correct Answer)

Consider a circular disc of radius 20 cm with centre located at the origin. A circular hole of radius 5 cm is cut from this disc in such a way that the edge of the hole touches the edge of the disc. The distance of centre of mass of residual or remaining disc from the origin will be

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(A) 1.5 cm (B) 2.0 cm (C) 0.5 cm (D) 1.0 cm

Q63. MCQ (Single Correct Answer)

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Refer to the circuit diagram given in the figure. which of the following observations are correct?

A. Total resistance of circuit is 6Ω

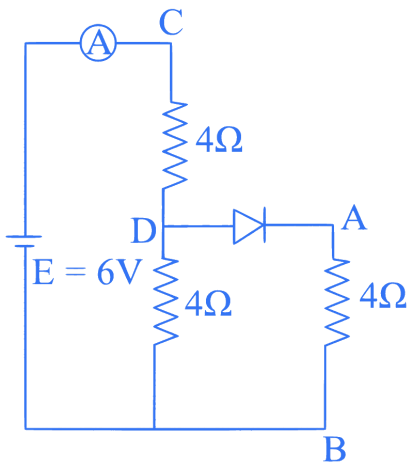
B. Current in Ammeter is 1 A

C. Potential across AB is 4 Volts.

D. Potential across CD is 4 Volts

E. Total resistance of the circuit is 8Ω.

Choose the correct answer from the options given below:



(A) B, C and E Only (B) A, C and D Only (C) A, B and C Only (D) A, B and D Only

Q64. MCQ (Single Correct Answer)

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The electric field of an electromagnetic wave in free space is $\overrightarrow{\mathrm{E}} = 57\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](4\hat{i}-3\hat{j})N/C$. The associated magnetic field in Tesla is

(A) $\overrightarrow{\mathrm{B}} = \frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](\hat{k})$

(B) $\overrightarrow{\mathrm{B}} = \frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](5\hat{k})$

(C) $\overrightarrow{\mathrm{B}} = -\frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](\hat{k})$

(D) $\overrightarrow{\mathrm{B}} = -\frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](5\hat{k})$

Q65. MCQ (Single Correct Answer)

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The electric flux is $\phi = \alpha\sigma + \beta\lambda$ where λ and σ are linear and surface charge density, respectively. $\left(\frac{\alpha}{\beta}\right)$ represents

(A) displacement (B) charge (C) electric field (D) area

Q66. MCQ (Single Correct Answer)

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Consider a moving coil galvanometer (MCG):

A. The torsional constant in moving coil galvanometer has dimensions $\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$

B. Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C. If we increase number of turns (N) to its double $(2\,\mathrm{N})$, then the voltage sensitivity doubles.

D. MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E. Current sensitivity of MCG depends inversely on number of turns of coil.

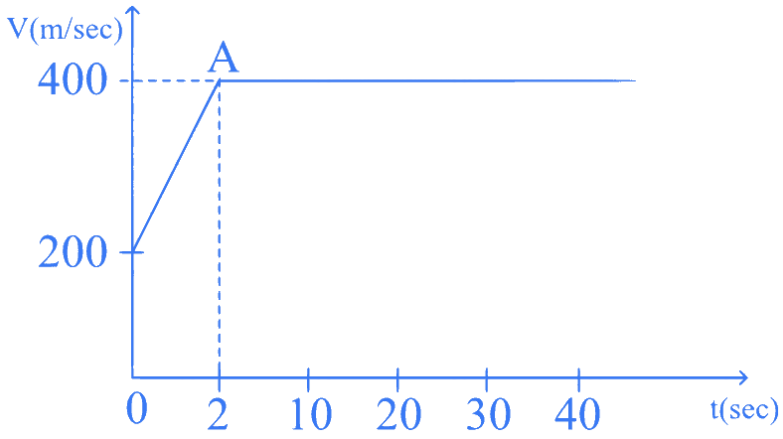
Choose the correct answer from the options given below:

(A) A, D Only **(B)** A, B Only **(C)** A, B, E Only **(D)** B, D, E Only

Q67. MCQ (Single Correct Answer)

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The motion of an airplane is represented by velocity-time graph as shown below. The distance covered by airplane in the first 30.5 second is_____ km .



(A) 12 **(B)** 6 **(C)** 3 **(D)** 9

Q68. MCQ (Single Correct Answer)

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Regarding self-inductance:

A. The self-inductance of the coil depends on its geometry.

B. Self-inductance does not depend on the permeability of the medium.

C. Self-induced e.m.f. opposes any change in the current in a circuit.

D. Self-inductance is electromagnetic analogue of mass in mechanics.

E. Work needs to be done against self-induced e.m.f. in establishing the current.

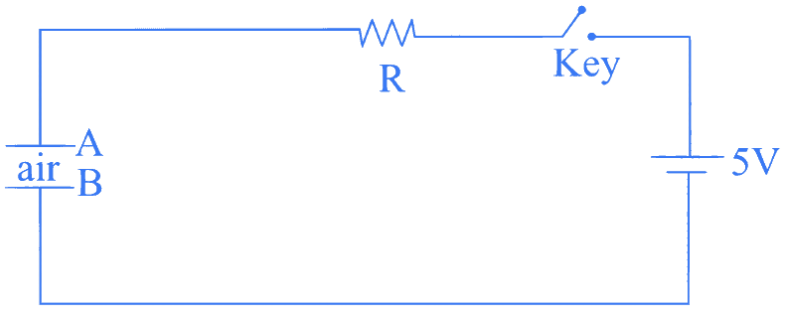
Choose the correct answer from the options given below:

(A) A, B, C, E only **(B)** A, B, C, D only **(C)** A, C, D, E only **(D)** B, C, D, E only

Q69. MCQ (Single Correct Answer)

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Identify the valid statements relevant to the given circuit at the instant when the key is closed.



A. There will be no current through resistor R.

B. There will be maximum current in the connecting wires.

C. Potential difference between the capacitor plates A and B is minimum.

D. Charge on the capacitor plates is minimum.

Choose the correct answer from the options given below:

(A) A, C Only **(B)** C, D Only **(C)** A, B, D Only **(D)** B, C, D Only

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Q70. MCQ (Single Correct Answer)

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Given a thin convex lens (refractive index μ_2), kept in a liquid (refractive index $\mu_1, \mu_1 < \mu_2$) having radii of curvatures $|R_1|$ and $|R_2|$. Its second surface is silver polished. Where should an object be placed on the optic axis so that a real and inverted image is formed at the same place?

(A) $\frac{(\mu_2+\mu_1)|R_1|}{(\mu_2-\mu_1)}$

(B) $\frac{\mu_1|\mathrm{R}_1|\cdot|\mathrm{R}_2|}{\mu_2(2|\mathrm{R}_1|+|\mathrm{R}_2|)-\mu_1\sqrt{|\mathrm{R}_1|\cdot|\mathrm{R}_2|}}$

(C) $\frac{\mu_1|R_1|\cdot|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_1|}$

(D) $\frac{\mu_1|\mathrm{R}_1|\cdot|\mathrm{R}_2|}{\mu_2(|\mathrm{R}_1|+|\mathrm{R}_2|)-\mu_1|\mathrm{R}_2|}$

Q71. Numerical

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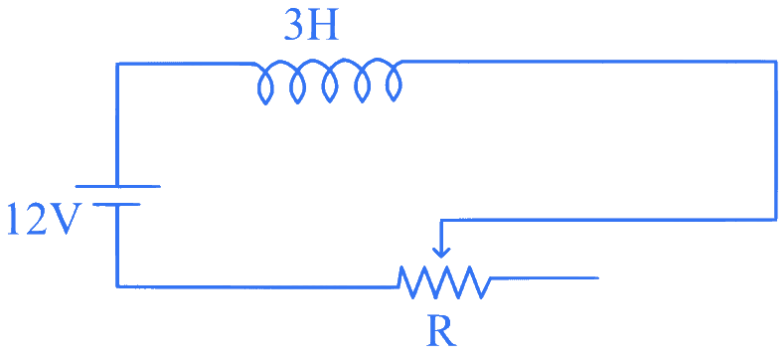
Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is _____ .

Q72. Numerical

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A force $\mathrm{f} = \mathrm{x}^2\mathrm{y}\hat{\mathrm{i}} + \mathrm{y}^2\hat{\mathrm{j}}$ acts on a particle in a plane $\mathrm{x} + \mathrm{y} = 10$. The work done by this force during a displacement from $(0, 0)$ to $(4\,\mathrm{m}, 2\,\mathrm{m})$ is _____ Joule (round off to the nearest integer)

Q73. Numerical



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In the given circuit the sliding contact is pulled outwards such that electric current in the circuit changes at the rate of $8\ \mathrm{A/s}$. At an instant when R is 12Ω, the value of the current in the circuit will be _____ A.

Q74. Numerical

A positive ion A and a negative ion B has charges $6.67 \times 10^{-19}\mathrm{C}$ and $9.6 \times 10^{-10}\mathrm{C}$, and masses $19.2 \times 10^{-27}\ \mathrm{kg}$ and $9 \times 10^{-27}\ \mathrm{kg}$ respectively. At an instant, the ions are separated by a certain distance r. At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $\mathrm{P} \times 10^{-13}$, where the value of P is _____.

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \mathrm{Nm^2C^{-2}}$ and universal gravitational constant as $6.67 \times 10^{-11}\mathrm{Nm^2\ kg^{-2}}$)

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Q75. Numerical

An ideal gas initially at $0^\circ\mathrm{C}$ temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is $3/2$, the change in temperature due to the thermodynamic process is _____ K.

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Answer Key

Chemistry

1. **D**	2. **C**	3. **B**	4. **A**	5. **D**
6. **A**	7. **C**	8. **C**	9. **B**	10. **A**
11. **B**	12. **C**	13. **D**	14. **D**	15. **D**
16. **D**	17. **C**	18. **B**	19. **D**	20. **A**
21. **897**	22. **2850**	23. **7**	24. **171**	25. **40**

Mathematics

26. **B**	27. **C**	28. **D**	29. **C**	30. **A**
31. **D**	32. **D**	33. **C**	34. **B**	35. **D**
36. **A**	37. **B**	38. **B**	39. **A**	40. **B**
41. **C**	42. **D**	43. **A**	44. **D**	45. **C**
46. **612**	47. **30**	48. **117**	49. **19**	50. **77**

Physics

51. **D**	52. **B**	53. **D**	54. **A**	55. **C**
56. **D**	57. **D**	58. **C**	59. **A**	60. **A**
61. **A**	62. **D**	63. **D**	64. **D**	65. **A**
66. **B**	67. **A**	68. **C**	69. **D**	70. **D**
71. **3**	72. **152**	73. **3**	74. **BONUS**	75. **273**

For numerical answers given as a range, any value within the range is correct.

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